Failure Testing of Material Specimens

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Section 11960 March 29, 2026

***Abstract*—This experiment used displacement-controlled uniaxial tensile and compressive testing to develop stress-strain relationships for an unknown metallic specimen, carbon fiber, nylon, and plaster of Paris, and to use those responses to determine material properties, compare failure behavior, and identify the unknown material. Testing was performed with an Instron 5967 universal testing machine, and, for the unknown metal, direct extensometer strain was used to improve stiffness and yield-property determination. The unknown specimen exhibited ductile metallic behavior with a modulus of 65.03 GPa, a 0.2% offset yield strength of 524.8 MPa, and an ultimate strength of 574.8 MPa, and comparison of its modulus, strength, elongation, and density indicated that it was most consistent with 7075-T6 aluminum. Among the tested materials, the carbon-fiber specimen had the highest specific strength and specific stiffness, the nylon specimen showed the greatest ductility and elongation before failure, and the plaster-of-Paris specimen showed the most brittle compressive response. A Ramberg–Osgood fit applied to the plastic region of the unknown-metal response gave $n = 0.086$ and $K \approx 778$ MPa with $R^2 = 0.9943$, indicating that the selected plastic-region data followed the model closely. Overall, the results clearly distinguished brittle and ductile material behavior and showed that combining stress-strain analysis, failure observations, uncertainty-supported property extraction, and plastic-model development can produce a defensible material identification.**

***Index Terms*— Aluminum alloy identification, Carbon fiber, Displacement-controlled testing, Ramberg–Osgood relationship, Stress-strain diagrams, Uncertainty analysis.**

I. INTRODUCTION

THIS experiment's goal was to use displacement-controlled uniaxial testing to create stress-strain diagrams for several materials. The data were then used to identify an unknown metallic specimen, compare material behavior, and determine material properties [1], [2]. An Instron model 2630-116 extensometer was provided for direct strain measurement when necessary, and an Instron 5967 Universal Testing Machine (UTM) with a 30 kN load cell was used for tensile and compressive testing [1], [2]. A carbon fiber tensile specimen, a nylon tensile specimen, an unknown metallic tensile specimen, and a plaster of Paris compressive specimen were all tested [1]. The measured load and deformation data were used to generate stress-strain curves for each specimen, from which modulus of elasticity, yield behavior, ultimate strength, breaking strength, percent elongation, toughness, specific strength, and specific stiffness could be determined [1], [4].

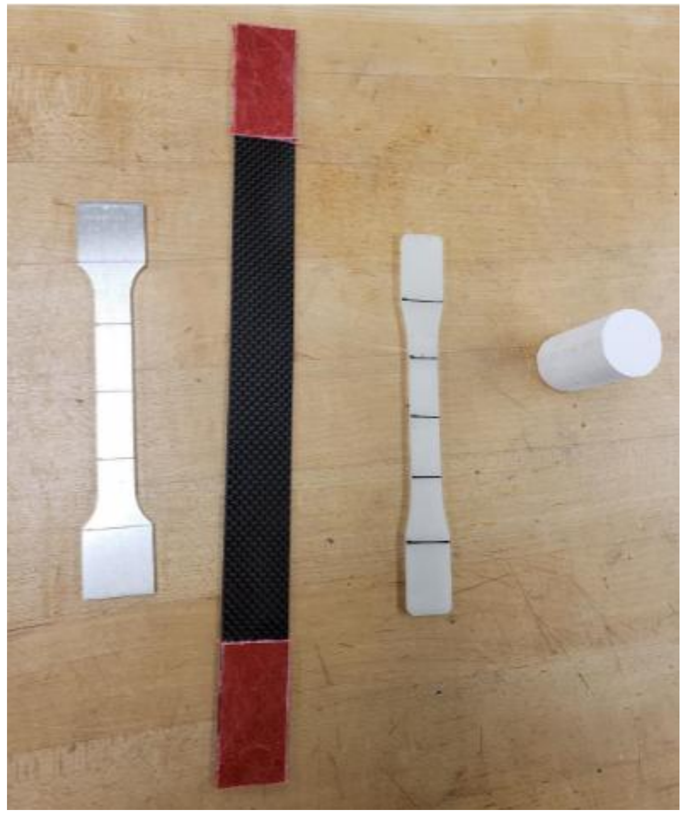

Fig 1. Test specimens used in Lab 3 before loading, including the unknown metal, carbon fiber, nylon, and plaster-of-Paris specimens from left to right [1].

The selected loading mode was determined by the specimen geometry as well as the anticipated material reaction. In order to immediately detect their elastic and, if any, plastic deformation from the stress-strain response, the unidentified metal and nylon specimens were examined under tension [1], [2]. Because it was anticipated that the plaster-of-Paris specimen would behave as a brittle compressive material, it was tested in compression [1], [2]. Tension tests were also conducted on the carbon-fiber specimen. Even though carbon fiber is brittle, non-axial phenomena like buckling would significantly affect a long, thin laminate loaded in compression, making it impossible for the test to accurately reflect a uniaxial material response [2]. Failure location and load transmission were also impacted by geometry. The unknown metallic specimen used a reduced gage section so that deformation and failure would occur away from the grips, while the carbon-fiber specimen used tabs to reduce grip-induced damage and improve force transfer into the laminate [2], [5]. The nylon specimen was prepared as a ductile tensile specimen with burnish marks for post-failure elongation measurements, and the plaster-of-Paris specimen used a cylindrical geometry suited for compression [1], [2].

An Instron 5967 Universal Testing Machine (UTM) with a 30 kN load cell was utilized for testing, and when direct strain measurement was necessary, an Instron 2630-116 extensometer was employed [1], [2]. The crosshead displacement was specified, and the associated load response was measured since the machine was run under displacement control. This was suitable for the lab since the goal was not just to record a failure load but also to generate complete stress-strain responses and study elastic, plastic, and failure behavior. The extensometer was used to measure strain directly for the unidentified metallic item. Instead, strain was calculated using measured displacement and specimen geometry for the carbon-fiber, nylon, and plaster-of-Paris specimens [1], [2], and [4]. The measurement chain in this experiment therefore consisted of the UTM, load cell, and either extensometer or crosshead-displacement output, followed by reduction of the recorded data into stress-strain form and then into material properties.

The primary reduced quantities in this experiment were engineering stress and engineering strain. Engineering stress, σ, was computed from the applied load, P, and the original cross-sectional area, A_o, as given in (1) [5].

$$\sigma = \frac{P}{A_o} \tag{1}$$

In (1), σ is engineering stress in Pa, P is the applied load in N, and A_o is the original specimen cross-sectional area in m^2. Engineering strain, ε, was computed from the change in specimen length, Δl, relative to the original length, l_o, as given in (2) [5].

$$\varepsilon = \frac{\Delta l}{L_o} \tag{2}$$

In (2), $\varepsilon$is engineering strain in mm/mm, $\Delta l$is the change in length in mm, and $l_o$is the original gage length in mm. These are engineering definitions because they use the original specimen dimensions rather than the instantaneous dimensions during deformation [5]. In this lab, the unknown metal used direct extensometer strain, while the remaining specimens required displacement-based strain estimation [1], [2], [5].

Within the initial linear portion of the stress-strain response, the modulus of elasticity, E, was determined from the slope of the elastic region, as given in (3) [2], [4], [5].

$$E = \frac{\sigma}{\varepsilon} \tag{3}$$

E stands for the modulus of elasticity in Pa, σ for stress in Pa, and ε for strain in mm/mm in (3). Only the roughly linear elastic range is covered by this relationship [2]. The lab also needed to determine the ultimate strength, breaking strength, yield strength when discernible, and 0.2% offset yield strength where necessary from the stress-strain curves [1], [4], and [5]. In the case of ductile materials, ultimate strength is the highest engineering stress attained before necking or ultimate failure, whereas yield denotes the change from mostly elastic to permanent deformation. The ultimate and failure strengths may occur at almost the same location for brittle materials, and yield may not be noticeable [4], [5].

Energy absorption and density-normalized performance were assessed using the entire stress-strain response in addition to modulus and strength measurements. According to (4) [1], [4], [5], toughness, or U_T, is the energy absorbed per unit volume up to failure and is equivalent to the area under the stress-strain curve.

$$U_T = \int \sigma d\varepsilon \tag{4}$$

In (4), $U_T$is toughness in J/m^3, $\sigma$is stress in Pa, and ε is strain in mm/mm. The lab also required comparison of specific stiffness and specific strength, which normalize stiffness and strength by material density so that materials with different masses may be compared on a more meaningful basis [1], [4], [5]. These quantities are given in (5) and (6).

$$Stiffness = \frac{E}{\rho} \tag{5}$$

$$Strength = \frac{\sigma}{\rho} \tag{6}$$

In equations (5) and (6), σ is the chosen strength quantity in Pa, E is the modulus of elasticity in Pa, and ρ is density in kg/m3. According to the lecture notes, the selection of strength must be supported by the material response. For instance, ultimate strength is better suited for brittle materials, but yield strength is often more suited for ductile materials [5]. This is particularly crucial when comparing the carbon fiber specimen to the metallic and polymeric specimens, since the ranking based just on strength may differ greatly from normalized performance [1], [4], [5].

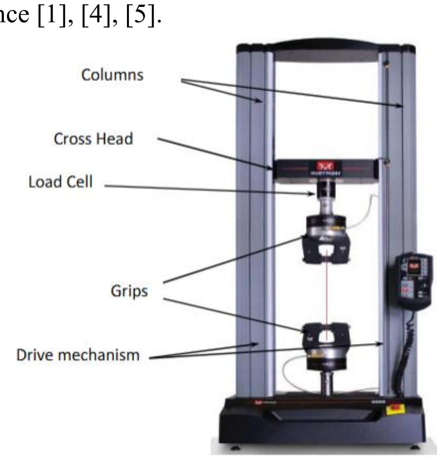

Fig 2. Instron 5967 UTM used for displacement-controlled testing, with the lo*ad path and extensometer setup shown for the unknown metal specimen [1].*

$$\Delta l_s = \Delta l - \Delta l_M = \Delta l - \frac{P}{K_M} \tag{7}$$

The idealized tensile stress-strain response utilized to analyze the ductile specimens in this experiment is depicted in Figure 3. The elastic region is represented by the first linear section, where the modulus of elasticity is determined by the slope. The beginning of yielding is indicated by the point where the curve starts to diverge from linearity; for materials without a distinct yield point, this can be determined using an offset method. In this lab, a 0.2% offset method was used for the unknown metal, while a 2% offset method was used for nylon because its transition from elastic to plastic response was more gradual [1], [3], [4]. Following yielding, the curve moves into the plastic region, strain hardens further, reaches its peak strength, and then starts to decline in the direction of the eventual breaking strength. For the unknown metal, the plastic portion of the response was later converted to true stress and true strain and used to develop a Ramberg–Osgood material model [4]. In addition, percent elongation was evaluated by both the final valid strain-based response and the change in burnished gage length after fracture, so that the recorded deformation could be compared to the permanent deformation visible in the failed specimen [1], [3], [4].

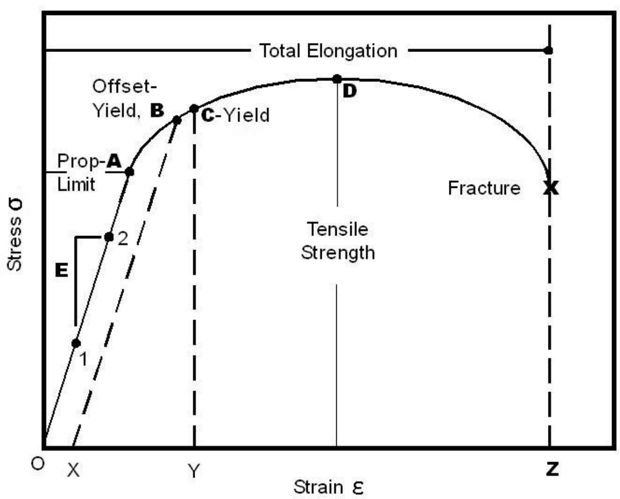

Fig 3. Idealized ductile engineering stress-strain curve showing the elastic region, yielding, plastic region, ultimate strength, and breaking strength, along with the offset-yield concept used in this lab [13].

P is the applied load in N, K_m is the machine stiffness in N/mm, Δls is the specimen deformation in mm, Δl is the total measured displacement in mm, and Δlm is the machine deflection in mm in (7). The machine stiffness with grips for this lab was specified at 75 kN/mm [5]. When machine deflection and specimen deformation are of the same order of magnitude, this correction applies to compressive specimens and tensile specimens 2 and 3 [5]. Without this adjustment, strain might be overstated, which would immediately lower the apparent modulus and change the stress-strain curve's form.

Additionally, the experiment required the identification of the unknown metallic specimen by applying uncertainty-supported reasoning to compare measurements to candidate material data [1], [4]. According to the discussion material, yield strength, breaking strength, ultimate strength, and elongation at failure can be used to refine the identification, while density and modulus can be employed initially to reduce the general material family [4]. A material candidate is only supported when the measured property range overlaps the reference property range in a way that can be quantitatively defended, according to the comparison standard, which was the 95% confidence interval [4]. This means that rather than being a distinct computation, uncertainty becomes crucial to the interpretation of the findings. If more than one candidate falls within the 95% interval for multiple measured properties, then the experiment may not have sufficient resolution to identify a single material definitively [1], [4].

Overall, this experiment related directly observed load and deformation data to elastic response, yielding, failure behavior, energy absorption, and density-normalized performance across four distinct specimen types using displacement-controlled uniaxial testing [1], [2], and [4]. The lab provided a foundation for material characterization and uncertainty-supported identification of the unknown metal by combining direct extensometer strain for the unknown metal with displacement-based strain estimation and machine-compliance correction for the remaining specimens [1], [2], [4], [5].

II. Procedure

A. Pre-Test Measurements

Each specimen's geometry was measured and documented prior to testing in order to compute stress, strain, and density later on. A plaster of Paris compressive specimen, a carbon fiber tensile specimen, a nylon tensile specimen, and an unidentified metallic tensile specimen were all tested. In order to get the mean dimensions and a 95% confidence interval, the cross-sectional dimensions of the unknown metal specimen were measured 10 times. In order to use their dimensions in the subsequent study, the remaining specimens were likewise measured prior to testing. Before loading started, the specimens of carbon fiber, nylon, plaster of Paris, and an unidentified metal were all recorded. When necessary for further density-based property computations, specimen mass was also noted.

B. Burnishing and Failure Measurements

The tensile specimens that were anticipated to experience noticeable plastic deformation were marked with burnishes prior to testing. In order to assess elongation following fracture, these markers were placed within the reduced portion of the specimen at intervals of one inch. Following failure, the broken specimen was put back together, and the final distance over a segment that had been two inches long was measured once more. This was utilized to calculate the elongation at failure based on the specimen's permanent deformation. In addition to these measures, each specimen's surface characteristics and fracture characteristics were recorded both during and after testing in order to later examine the failure mode.

C. Tensile Testing

After the specimens were measured, the lab instructor tested the tensile specimens using the Instron universal testing equipment. First to be loaded was the unidentified metallic specimen. In order to quantify strain immediately during the test, an extensometer was affixed to the unidentified specimen. After then, the specimen was subjected to tension loading at a steady crosshead speed of 7 mm/min until it failed. The specimen was removed after the fracture, and the burnish marks were used to assess the post-failure extension.

The carbon fiber tensile specimen was then mounted in the machine using its tabs to reduce grip-related damage. This specimen was also tested at a crosshead speed of 7 mm/min until failure. Since the extensometer was not used for this specimen, its strain was later determined from displacement-based deformation rather than direct extensometer data.

The nylon samples underwent tension testing last. The crosshead speed at the start of the test was 7 mm/min. In order to finish the large-deformation section of the test more quickly, the speed was increased after the reaction stabilized. After that, the specimen was loaded until it failed or the machine's trip limit was reached. The final elongation was measured using the burnish marks following testing, much like with the unidentified metal specimen.

D. Compression Testing

After the tensile tests were completed, the testing machine was reconfigured for compression. The tensile grips were removed and replaced with the compression setup. The plaster of Paris specimen was centered in the compression fixture and loaded until failure while the machine recorded the test response. Since an extensometer was not used for this specimen, strain was later estimated from the measured displacement and the original specimen height.

E. Recorded Data

The machine's output data was retained for later analysis during testing. All specimens' load and displacement were included in these data, along with strain information from the extensometer when it was available. The extensometer strain was applied directly to the unidentified metallic object. Strain for the carbon fiber, nylon, and plaster of Paris specimens was calculated later using measured displacement and specimen dimensions rather than immediately from the extensometer.

F. Data for Analysis

For further reduction into stress-strain form, the recorded geometry measurements, mass data, machine outputs, and post-failure elongation measurements were kept. The necessary material parameters, such as modulus of elasticity, yield or offset yield strength when appropriate, ultimate strength, breaking strength, percent elongation, toughness, specific strength, and specific stiffness, were ascertained using these numbers. In order to more accurately estimate specimen deformation for the specimens that employed displacement-based strain, machine compliance was considered later in the analysis.

III. Results

Engineering stress-strain diagrams were created for each specimen based on the direct measurements made in the lab and the load-displacement data gathered by UTM. Stress-strain graphs and the measured specimen dimensions are shown in this section. In Section IV, the retrieved material-property tables are displayed. The specimen spreadsheet's statistically derived measurement uncertainty is included in the unknown-metal subsection; in accordance with lab guidelines, the remaining direct measurements are simply presented with instrument uncertainty.

A. Unknown Specimen

Before testing, the unknown metallic specimen's initial width and thickness were measured ten times. The average thickness was 3.1469 ± 0.0015 mm, while the average width was 10.287 ± 0.023 mm. An average initial cross-sectional area of 32.372 ± 0.073 mm^2 was obtained from these measurements. The mass of the specimen was 17.12 ± 0.005 g. The specimen volume was 7.21 ± 0.14 cm^3 based on the specimen geometry. The initial 50.8 ± 0.5 mm burnished section was measured as 60.96 ± 0.5 mm following tensile failure. Table I lists the individual width and thickness measurements that were utilized to calculate the mean dimensions.

TABLE I
UNKNOWN REPEATABILITY MEASUREMENTS

Width (mm)	Thickness (mm)
10.23 ± 0.0025	3.152 ± 0.0025*
10.33 ± 0.0025	3.146 ± 0.0025
10.29 ± 0.0025	3.145 ± 0.0025
10.32 ± 0.0025	3.148 ±0.0025
10.25 ± 0.0025	3.147 ± 0.0025
10.27 ± 0.0025	3.147 ± 0.0025
10.28 ± 0.0025	3.145 ± 0.0025
10.32 ± 0.0025	3.145 ± 0.0025
10.29 ± 0.0025	3.148 ± 0.0025
10.29 ± 0.0025	3.1460.0025

** Indicates that value was found to be an outlier.*

Fig. 4 displays the engineering stress-strain response for the unidentified metallic material. After rising through the elastic area and reaching its maximum engineering stress, the curve began to decline in the direction of the ultimate breaking stress.

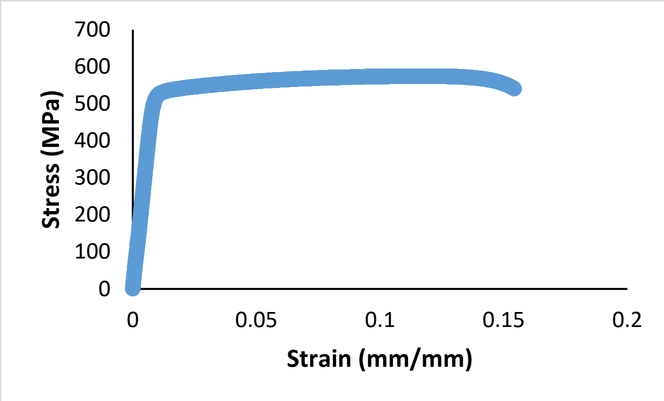

Fig 4. Engineering stress-strain diagram for the unknown metallic *specimen.*

B. Carbon Fiber Specimen

The original dimensions of the carbon-fiber tensile specimen were 25.555 ± 0.025 mm for width and 0.817 ± 0.0025 mm for thickness. With these measurements, the initial cross-sectional area was 20.878 mm^2. The initial distance between the grips was 206 ± 0.5 mm, and the measured specimen mass was 18.84 ± 0.005 g.

The engineering stress-strain response for the carbon-fiber specimen is shown in **Fig. 5**. The measured stress increased with strain until failure, and no extended plastic region was visible in the recorded response.

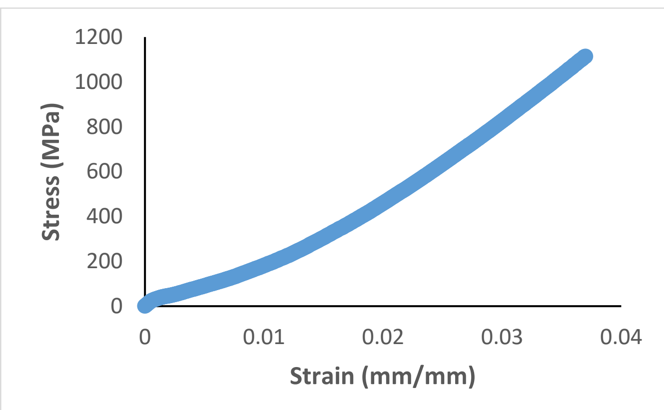

Fig 5. Engineering stress-strain diagram for the carbon-fiber specimen.

C. Nylon Specimen

The original dimensions of the nylon tensile specimen were 12.636 ± 0.0025 mm for width and 3.045 ± 0.0025 mm for thickness. With these measurements, the original cross-sectional area was 38.477 mm^2. 104 ± 0.5 mm was the initial distance between the grips. The initial 50.8 ± 0.5 mm burnished section was remeasured as 101.6 ± 0.5 mm following testing. For the subsequent strain and elongation computations, these values were kept.

The engineering stress-strain response for the nylon specimen is shown on **Fig. 6**. The nylon specimen exhibited the largest tensile strain range of the tested specimens before failure.

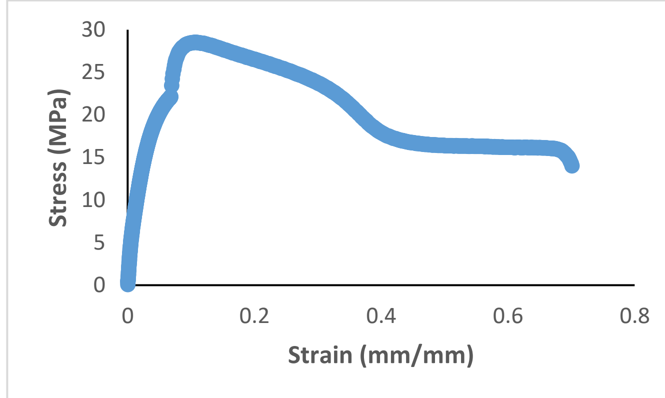

Fig 6. Engineering stress-strain diagram for the nylon specimen.

D. Plaster of Paris Specimen

The dimensions of the plaster-of-Paris compressive specimen were 25.659 ± 0.025 mm in diameter and 54.051 ± 0.025 mm in height. With these measurements, the original cross-sectional area was 517.082 mm^2. The compressive stress-strain response was produced using these parameters.

The engineering stress-strain response for the plaster-of-Paris specimen is shown in **Fig. 7**. The compressive stress increased with strain until failure and did not exhibit the extended post-yield deformation observed in the ductile tensile specimens.

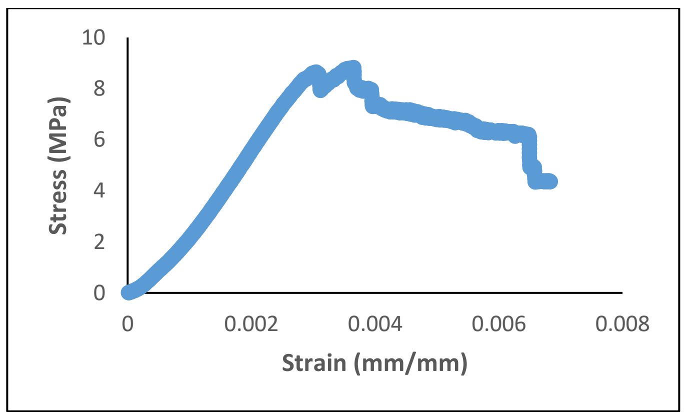

Fig 7. Engineering stress-strain diagram for the plaster-of-Paris specimen.

IV. Discussion

The objective of this experiment was to use displacement-controlled uniaxial testing to develop stress-strain relationships for the unknown metal, carbon fiber, nylon, and plaster-of-Paris specimens and then use those curves to determine the required material properties [1], [2]. The measured responses were physically consistent with the expected differences between ductile metals and polymers, brittle composite behavior, and brittle compressive failure. For the unknown metallic specimen, the measured properties and their 95% uncertainty were also used to support material identification rather than relying on a single nominal value [1], [4], [5].

Use The stress-strain curves were used to directly extract material properties. To prevent slack from affecting the response at the start of the test, the elastic modulus was calculated from the linear portion of each curve using a linear fit over the central part of the elastic region. For the unknown specimen, the linear fit used to determine E was applied over the stress range from 0 to 600 MPa. Only the ductile tensile specimens had their yield strength measured. Because the nylon specimen's elastic-to-plastic transition was more gradual, a 2% offset approach was applied, while a 0.2% offset method was utilized for the unknown metal [1], [4], [5]. Brittle materials such as the carbon-fiber laminate and the plaster-of-Paris specimen were not assigned to yield strength because their curves did not show a clear yielding region before failure.

Breaking strength was defined as the engineering stress at ultimate failure, whereas ultimate strength was defined as the maximum engineering stress on the stress-strain curve. Because the cross-sectional area started to localize after the maximum engineering stress was reached, these values for the ductile materials happened at different times. Because there was very little stable post-peak deformation prior to fracture, the ultimate and failure strengths for the brittle specimens were significantly closer. The area under the stress-strain curve up to failure was used to calculate toughness, and the recorded strain history and the change in the burnished gauge length following failure were used to calculate % elongation [1], [4], and [5]. Density was then used to normalize the strength and stiffness values so that specific strength and specific stiffness could be compared across materials with very different masses [4], [5].

The final reported values were used to do the entire uncertainty analysis for the unknown specimen. The propagation process described in the Appendix was used to combine statistical uncertainty from the repeated width and thickness measurements with the provided instrument uncertainties. Therefore, rather than relying just on nominal values, comparisons involving the unknown specimen should be based on the 95% ranges. This is particularly crucial for the material-identification process because, when only one feature is considered, multiple candidate alloys may seem comparable [1], [4], [5].

Direct extensometer strain should be used whenever the specimen deformation range allows it because it measures specimen strain directly and reduces error from machine compliance and grip seating. In this experiment, which made the unknown-metal modulus and yield calculations more reliable than the displacement-based strain calculations used for the carbon-fiber, nylon, and plaster-of-Paris specimens. Crosshead displacement is still useful when the extensometer cannot be used safely or would exceed its operating range, but it should be interpreted more cautiously because the recorded motion includes both specimen deformation and machine deflection.

A. Unknown Metal Specimen

The final property values should be given in Table II, and the 0.2% offset line for the unknown metallic specimen should be displayed in Fig. 8. The unknown specimen's reported elastic modulus of 65.0 GPa immediately put it much below steel and far closer to the aluminum-alloy range. Both a visual yield estimate and a 0.2% offset yield estimate were appropriate for this specimen because the same curve also demonstrated a distinct elastic region followed by a plastic region. The start of yielding was consistently observed if the visual and offset yield values differed just little and their 95% intervals overlapped.

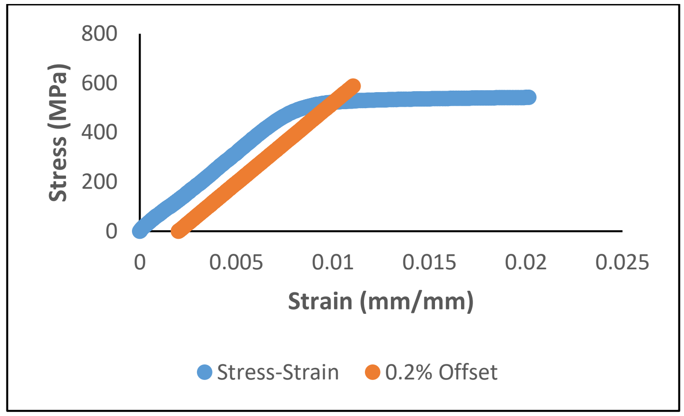

Fig 8. Stress strain curve and 0.2% offset line for unknown metal.

TABLE II
MATERIAL PROPERTIES FOR UNKNOWN SPECIMEN

Symbol	Value	Units
E	65.03 ± 0.02	GPa
σ_y (.2%)	524.8 ± 2.88	MPa
σ_y (Visual)	497.8 ± 3.05	MPa
σ_u	574.8 ± 3.15	MPa
σ_b	322 ± 1.77	MPa
Toughness	87.04	MJ/m^3
$\%EL_1$	15.96 ± 0.39	%
$\%EL_2$	20.0 ± 0.08	%
Density	2.374 ± 0.047	g/cm^3
Specific Strength	221 ± 1.21	MPa*cm^3/g
Specific Stiffness	27.39 ± 0.55	GPa*cm^3/g

The unknown specimen's reported density of 2.37 g/cm3 supports the aluminum family instead of steel or copper-base alloys. The specimen is more consistent with a high-strength aluminum alloy than a low-strength commercial alloy when this density is considered along with the measured modulus, 0.2% offset yield strength, and ultimate strength. The unknown should not be classified as a general-purpose alloy like annealed or lightly reinforced aluminum because the measured strength values are significantly higher than those anticipated for typical low-strength aluminum grades. Identification of the alloy should be done carefully and only after comparing the recorded 95% intervals to the chosen candidate material ranges because the measured density is lower than several wrought aluminum alloys.

It was evident that the unidentified metal specimen was ductile. The specimen experienced apparent permanent deformation in the smaller portion prior to fracture, as illustrated in Fig. 9, and the two parts can be reassembled along an inclined fracture plane as opposed to a flat, brittle separation.

In line with a tensile specimen that first yielded, then accumulated plastic strain, and failed by shear-dominated fracture following localization, the fracture surface was not perpendicular to the loading direction. Since the gage section lengthened permanently prior to rupture rather than failing abruptly with very little force, the increased spacing between the burnish marks following failure supports the same conclusion. This failure shape agrees with the stress-strain curve, which showed a clear elastic region, a plastic region, an ultimate stress, and then a drop toward the final breaking stress.

Fig 9. Unknown specimen failure after tensile test.

Here, the trends in uncertainty are equally significant. Uncertainty in the repeated width and thickness measurements directly affected the yield, ultimate, and breaking stresses since the stress values were computed by dividing the load by the area. Due to the very tiny scale uncertainty, the density estimate was more sensitive to the geometric volume approximation than to the mass measurement itself. Unlike the displacement-based strain estimates used for the other specimens, the modulus calculation benefited from direct extensometer strain, which reduced the stiffness estimate's sensitivity to machine-compliance effects. For this reason, one of the best characteristics for identifying the unknown specimen's material family is its modulus.

B. Carbon Fiber Specimen

The behavior of the carbon-fiber material was significantly different from that of the unidentified metal. Its stress-strain response did not exhibit a discernible yield area and remained linear until failure, which is consistent with brittle tensile behavior in a fiber-dominated laminate. That interpretation is supported by the failure image in Fig. 10. The specimen failed suddenly close to the tabbed end rather than exhibiting necking or significant plastic deformation. Instead of a neat break centered in a long gauge region, the fractured region exhibits fragmented fibers, local separation, and a failure point near the tab transition. This is significant because failure close to the tab or grip might result in stress concentrations and lower the reported strength below the laminate's entire tensile capacity.

Fig 10. Failed carbon-fiber specimen showing fracture near the tabbed ends.

Table III lists the measured carbon-fiber characteristics. The ultimate strength was 1113.88 MPa, the final breaking strength was 165.27 MPa, and the modulus of elasticity was 37.8 GPa. 8.93 MJ/m^3 was the measured toughness. The specimen density was 1.44 g/cm^3, the specific strength was 775.1 MPa·cm^3/g, and the specific stiffness was 26.3 GPa·cm^3/g. The specimen had the highest specific strength and specific stiffness of the materials tested, which is physically compatible with a low-density composite loaded along stiff load-carrying fibers, even though it failed prematurely near the tabbed region.

TABLE III
MATERIAL PROPERTIES FOR CARBON FIBER

Symbol	Value	Units
E	37.8	GPa
σ_u	1113.88	MPa
σ_b	165.27	MPa
Toughness	8.93	MJ/m^3
Density	1.44	g/cm^3
Specific Strength	775.1	MPa*cm^3/g
Specific Stiffness	26.3	GPa*cm^3/g

Yield strength was not reported for the carbon-fiber specimen because no distinct yielding region was observed before failure.

Compared to the published mechanical-property data for carbon-fiber/epoxy composite materials, the tested carbon-fiber specimen showed lower measured stiffness and strength than would be expected for an ideal laminate. That comparison should be interpreted cautiously, however, because the reference source reports general fibre/epoxy composite behavior across different fiber orientations, while the specimen tested here failed near the tabbed end rather than fully within the gauge section. As a result, the measured values in this experiment are more representative of the tested coupon and its failure location than of the maximum possible performance of the composite material itself. [14]

C. Nylon Specimen

Among the materials studied, the nylon specimen exhibited the highest degree of ductility. Among the tensile specimens, its stress-strain curve had the lowest beginning slope, but it went through the greatest range of strains before failing. That behavior is seen in the failed specimen depicted on Fig. 11. The gauge section narrowed in the deformed area and stretched significantly and only ruptured after a significant visible elongation. This failure structure is in line with the behavior of ductile polymers, which can experience significant plastic deformation and local drawing before ultimate separation.

Fig 11. Failed nylon specimen after tensile testing.

The measured nylon properties are listed in Table IV. The modulus of elasticity was 0.886 GPa, the 2% offset yield strength was 18.4 MPa, the ultimate strength was 28.51 MPa, and the breaking strength was 14.00 MPa as seen in Figure 12. The measured toughness was 14.23 MJ/m3. The two elongation values were 70.11% and 69.23%, which are both much larger than the corresponding values for the unknown metal. The density was 1.14 g/cm3, which gave a specific strength of 10.49 MPa·cm3/g and a specific stiffness of 0.777 GPa·cm3/g. These results are physically consistent with a material that is compliant and highly ductile rather than stiff and strong.

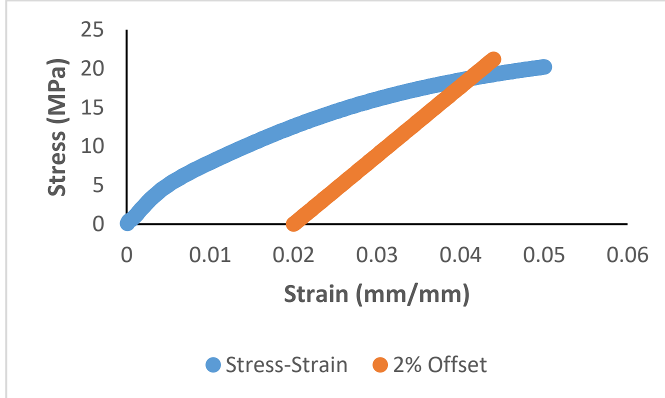

Fig 12. Stress strain curve and 02% offset line for nylon specimen.

TABLE IV
MATERIAL PROPERTIES FOR NYLON SPECIMEN

Symbol	Value	Units
E	0.886	GPa
σ_y (2%)	18.4	MPa
σ_y (Visual)	19.3	MPa
σ_u	28.51	MPa
σ_b	14.00	MPa
Toughness	14.23	MJ/m^3
$\%EL_1$	70.11	%
$\%EL_2$	69.23	%
Density	1.14*	g/cm^3
Specific Strength	10.49	$MPa*cm^3/g$
Specific Stiffness	0.777	$GPa*cm^3/g$

** Indicates value was sourced from [8]*

Because the nylon's strain was calculated using displacement-based deformation of direct extensometer output, the results should still be evaluated cautiously. To finish the large-deformation section of the test more quickly, the crosshead speed was also raised. The apparent yield behavior and strain at failure may have been impacted by that speed variation since polymers are strain-rate sensitive. Despite these drawbacks, the observed failure form and the measured reaction both confirm that nylon was the most ductile specimen examined.

Compared to the selected reference nylon values, the tested specimen had lower stiffness and lower strength, but a larger elongation prior to failure. That trend is physically reasonable for a specimen that experienced large plastic deformation and local drawing in the gauge region before rupture. The difference is also consistent with the testing method used here, since nylon strain was estimated from displacement-based deformation and the crosshead speed was increased during the test, both of which can affect the measured modulus, yield behavior, and strain at failure. For that reason, the reference values are best used as a comparison baseline rather than as an exact target that the measured specimen was expected to match. [8]

D. Plaster of Paris Specimen

The specimen made of plaster-of-Paris exhibited brittle compressive behavior. Without the prolonged deformation observed in the ductile tensile specimens, its stress-strain curve increased to a maximum compressive stress before declining. That conclusion is supported by the failed specimen image in Fig. 13. Instead of expanding or flowing plastically, fragments chipped away from the edge of a visible fissure that ran the length of the cylinder. A cast, porous, gypsum-based material failing by local cracking and crushing under compression is consistent with that type of response.

Fig 13. Failed plaster-of-Paris specimen after compressive testing.

The measured plaster-of-Paris properties are listed in Table V. The compressive modulus was 3.13 GPa, the ultimate compressive stress was 8.82 MPa, and the final breaking stress was 6.10 MPa. The measured toughness was 0.0385 MJ/m^3. Using the approved external density source, the density was 2.32 g/cm^3, which gave a specific strength of 3.80 MPa·cm^3/g and a specific stiffness of 1.35 GPa·cm^3/g. These values are all low relative to the metallic and composite specimens, which are consistent with the observed brittle behavior and the nature of a cast plaster specimen. Yield strength was not reported for plaster of Paris because the specimen failed in brittle compression without a discernible yield region.

TABLE V
MATERIAL PROPERTIES FOR PLASTER OF PARIS

Symbol	Value	Units
E	3.13	GPa
σ_u	8.82	MPa
σ_b	6.10	MPa
Toughness	0.0385	MJ/m^3
Density	2.32*	g/cm^3
Specific Strength	3.80	MPa*cm^3/g
Specific Stiffness	1.35	MPa*cm^3/g

** Indicates value was sourced from [9]*

Plaster of Paris, in contrast to the metal specimen, is extremely sensitive to cure conditions, moisture content, trapped porosity, and casting quality. Therefore, rather than being definite material constants, the measured values are best understood as indicative of the tested specimen. This specimen was valuable in the lab mostly because it demonstrated the differences between the more ductile tensile responses and brittle compression failure.

E. Material Comparison

When the materials are compared as a group, the trends in the measured properties are physically consistent. The unknown metal had the highest absolute modulus at 65.03 GPa, followed by the carbon-fiber specimen at 37.8 GPa, the plaster-of-Paris specimen at 3.13 GPa, and the nylon specimen at 0.886 GPa. That ordering agrees with the expected difference between metals, structural composites, cast brittle materials, and ductile polymers. The unknown metal also combined high stiffness with substantial ductility, while the carbon-fiber specimen combined lower absolute stiffness with much better density-normalized performance.

Low-density composites are very helpful when mass is crucial because the carbon-fiber specimen has the highest specific strength and specific stiffness. The nylon specimen was the most ductile material in the collection because it exhibited the greatest elongation prior to failure. The plaster-of-Paris specimen responded the most brittlely and performed the worst overall. These comparisons are important because they demonstrate that rather than opposing one another, the computed characteristics, failure morphologies, and stress-strain curves all support the same physical explanation.

F. Unknown Material Identification

A comparison chart containing the measured modulus, yield strength, ultimate strength, elongation, and density should be used to identify the unknown specimen. Modulus and density are the most crucial first-step characteristics since they distinguish large material families more well than strength alone. In this instance, the measured density of 2.374 ± 0.047 g/cm^3 and modulus of 65.03 ± 0.02 GPa firmly support the aluminum family and exclude titanium alloys, copper alloys, and steels. The ultimate strength of 574.67 ± 3.15 MPa and the recorded 0.2% offset yield strength of 524.76 ± 2.88 MPa suggest that the unknown was a higher-strength aluminum alloy rather than a low-strength commercial aluminum.

TABLE VI
MATERIAL PROPERTY COMPARISON FOR UNKNOWN SPECIMEN

Parameter	Specimen	7075-T6	2024-T351	6061-T6	Units
E	65.03 ± 0.02	71.7	73.1	68.9	GPa
σ_u	574.8 ± 3.15	572	469	310	MPa
σ_y	524.8 ± 2.88	503	324	276	MPa
$\%EL_1$	15.96 ± 0.39	11	19	12	%
Density	2.374 ± 0.047	2.81	2.78	2.7	g/cm^3

The comparison in Table VI indicates that the unknown specimen was most consistent with 7075-T6 aluminum. Of the three candidate alloys, 7075-T6 gave the closest match to the measured yield and ultimate strengths. The specimen yield strength, 524.8 ± 2.88 MPa, was closest to the 7075-T6 value of 503 MPa, and the measured ultimate strength, 574.8 ± 3.15 MPa, was identical to the listed 572 MPa. Since the modulus values of the candidate alloys were close to one another, the strength values were more useful for separating the candidates after the material family had already been narrowed to aluminum.

Using the 95% uncertainty intervals, 6061-T6 was not a strong candidate because both its yield strength and ultimate strength were well below the measured specimen range. 2024-T4/T351 was closer in elongation, but its reported yield and ultimate strengths were still lower than the measured values. Of the candidate alloys considered, 7075-T6 provided the closest agreement in both $\sigma_y$and σ_u, which made it the most consistent choice after the material family had already been narrowed to aluminum.

For a high-strength aluminum alloy, the elongation result made sense as well. The measured elongation remained far more consistent with the aluminum-alloy group than with a lower-strength structural metal or a different material family, even though it did not precisely match one possibility. Density was the primary discrepancy in comparison. For all three candidate alloys, the measured density of 2.374 ± 0.047 g/cm^3 was lower than the specified values; nevertheless, density in this experiment was more sensitive to geometric uncertainty than strength values because it was dependent on the projected specimen volume. Because the modulus, yield strength, ultimate strength, and ductile fracture shape all support the same conclusion, the density difference is more reasonably attributed to uncertainty in the volume estimate than to the specimen being a different alloy family.

The unknown specimen was most consistent with 7075-T6 aluminum, or at the very least a high-strength aluminum alloy with characteristics extremely like 7075-T6, according to the complete comparison. The measured strength values, the stress-strain response, and the observed ductile failure morphology all corroborate this conclusion, making it more reliable than identifying the specimen based on modulus.

G. Ramberg–Osgood Plastic Model for the Unknown Metal

A Ramberg–Osgood plastic material model was created using the plastic section of the unknown-metal response. A fitting region was chosen from the post-yield, pre-ultimate part of the reaction after the engineering stress-strain data were transformed into true stress and true strain in accordance with the course procedure. The strength coefficient K and the strain-hardening exponent n were then calculated by plotting the chosen data in log-log form [5].

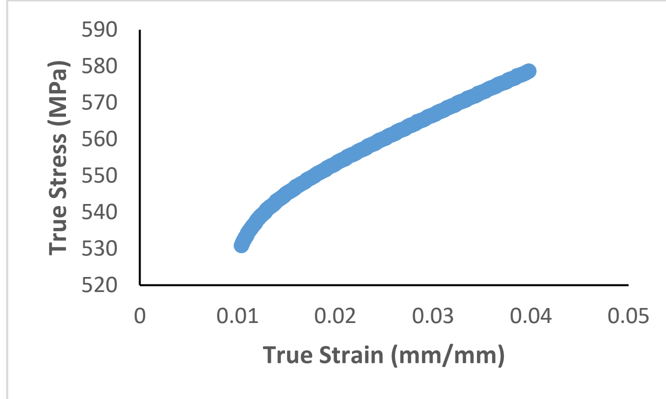

Fig 14. True stress–true strain curve for the plastic region of the unknown metall*ic specimen.*

The chosen plastic part of the true stress–true strain response is displayed in Figure 14. Rather than brittle failure or ideally plastic behavior, this region smoothly increased after yielding and before the specimen reached its maximum stress, indicating measurable strain hardening.

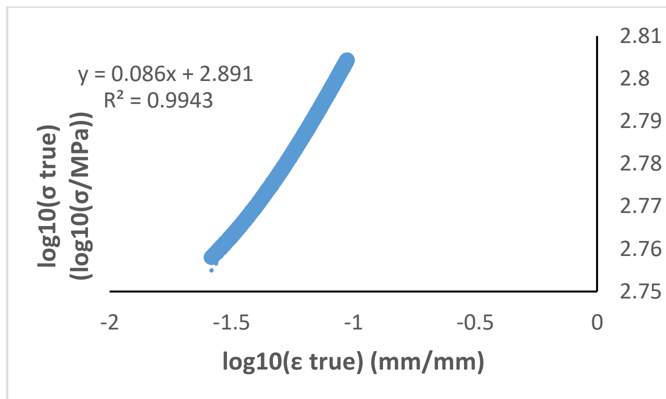

Fig 15. Log-log fits the plastic region of the unknown metallic specimen with linear trendline and R^2

As shown in Fig. 15, the log-log fit of the chosen plastic-region data yielded a strength coefficient of $K \approx 778$ MPa and a strain-hardening exponent of n = 0.086. These figures came from the trendline equation, where log10(K) is the intercept and n is the slope. The chosen plastic-region data closely matched the Ramberg–Osgood power-law form over the fitted range, as indicated by the fit's R^2 value of 0.9943 [5].

For a ductile strain-hardening metal, the fitted parameters make sense mechanically. While the value of K being higher than the recorded yield strength is likewise consistent with strengthening during plastic deformation, the positive value of n indicates that the specimen continued to harden after yielding rather than deforming as a perfectly plastic material. This plastic-model result demonstrates that the measured plastic reaction may be accurately represented by the necessary constitutive form, supporting the conclusion that the unknown material was 7075-T6 aluminum. Instead of providing a comprehensive explanation of the entire stress-strain curve, the fit should still be understood as a model for the chosen plastic region.

H. Limitations and Improvements for Material Testing

Uncertainty in the geometry-based calculations, particularly the density estimate for the unknown specimen, was the experiment's primary drawback. The density result would be better and the alloy-identification step would be more convincing with a more precise volume measurement. Another drawback was that repeatability could not be properly assessed because only one specimen of each material was tested.

Future testing would be strengthened by a few enhancements. Stiffness calculations might be less unclear if direct extensometer strain were used whenever feasible. Repeatability would be enhanced by testing many specimens of each substance. Enhancing the tab and grip transition would lessen early end failure in carbon fiber. For nylon, maintaining a constant crosshead speed would make the results easier to compare to reference data. Tighter control over casting and curing would lessen the variability caused by moisture and porosity in plaster of Paris.

Overall, the experiment generated results that were physically consistent with each material class and clearly distinguished between brittle and ductile behavior across the investigated materials. The mechanical characteristics and ductile metallic reaction of the unidentified specimen were most comparable with 7075-T6 aluminum; the density estimate, not the strength-based comparison, was the primary source of ambiguity.

V. Conclusion

Using stress-strain behavior, computed material parameters, and observed failure modes, this work used displacement-controlled tensile and compressive tests to describe the mechanical response of an unknown metal, carbon fiber, nylon, and plaster of Paris. With a modulus of 65.03 GPa, yield strength of 524.8 MPa, and ultimate strength of 574.8 MPa, the unidentified specimen had ductile metallic behavior. A comparison of its modulus, strength, elongation, and density revealed that it was most comparable with 7075-T6 aluminum. The nylon specimen demonstrated the most ductility and elongation prior to failure, the plaster-of-Paris specimen displayed the most brittle compressive response, and the carbon-fiber specimen had the highest specific strength and specific stiffness among the investigated materials. The Ramberg–Osgood fit for the unknown metal also represented the selected plastic region well, which supported the measured strain-hardening behavior. Overall, the results clearly showed the difference between brittle and ductile behavior across the tested materials, and future improvements should focus on reducing geometry-based uncertainty, improving failure-location control, and testing multiple specimens of each material for better repeatability.

Appendix

Uncertainty Analysis

The uncertainty analysis utilized in this report is outlined in this appendix. As required by the lab instructions, full propagation was only carried out through the reported material properties of specimen 1, the unknown metal, but direct-measurement uncertainties were assigned to all measured quantities. The statistical analysis of the repeated width and thickness measurements, the propagation stages used to calculate the uncertainties in the reported unknown-metal properties, and the direct/given uncertainties are all included in the appendix. Uncertainty was not propagated for toughness or the Ramberg-Osgood plastic model, and propagated uncertainty was not generated for specimens 2-4 beyond direct measurements in accordance with the course guidelines [1], [5].

A. Direct and Given Uncertainties

Table VII lists the direct and given quantities used in the uncertainty analysis for specimen 1. These values were taken directly from the specimen sheet and the derived specimen-1 calculations.

TABLE VII
Direct and Given Uncertainties for Unknown Specimen

Name	Symbol	Uncertainty	Units	Source
Uncertainty of Width	μ_w	± 0.023	mm	95% CI [5]
Uncertainty of Thickness	μ_t	± 0.0015	mm	95% CI [5]
Uncertainty of Mass	μ_m	± 0.005	g	Rule of Thumb [5]
Uncertainty of Length Initial	μ_{Lo}	± 0.5	mm	Rule of Thumb [5]
Uncertainty of Length Final	μ_{Lf}	± 0.5	mm	Rule of Thumb [5]
Uncertainty of CAD Volume	μ_V	± 2%	Relative	Rule of Thumb [5]
Uncertainty of Area	μ_A	± 0.073	mm^2	RSS
Uncertainty of Young's Modulus	μ_E	± 0.02	GPa	Monte Carlo
Uncertainty of Micrometer	μ_{Mi}	± 0.0025	mm	Given [5]
Uncertainty of Caliper	μ_{Ca}	± 0.025	mm	Given [5]
Uncertainty of Ruler	μ_{Ru}	± 0.5	mm	Given [5]
Uncertainty of Scale	μ_{Sc}	± 0.005	g	Given [5]

B. Statistical Treatment of Width and Thickness

Ten measurements were made of the unknown specimen's thickness and width. The 95% confidence interval for each mean dimension was calculated using a t-distribution. Prior to calculating the final thickness uncertainty, one thickness measurement (3.152 mm) was found to be an outlier and eliminated. Table VIII lists the final preserved dimensions and confidence intervals. This was found using equations (8)-(13)

$$\bar{x} = \frac{1}{N} \cdot \sum_{i=1}^{N} x_i \quad (8)$$

$$S_x = \sqrt{\frac{1}{N-1} \cdot \sum_{i=1}^{N} (x_i - \bar{x})^2} \quad (9)$$

$$S_{\bar{x}} = \frac{S_x}{\sqrt{N}} \quad (10)$$

$$P_x = tS_x \quad (11)$$

$$CI = \bar{x} \pm P_{\bar{x}} \quad (12)$$

$$Bounds = \bar{x} \pm 2S_x \quad (13)$$

TABLE VIII
STATISTICAL UNCERTAINTIES FOR UNKNOWN SPECIMEN

Name	Symbol	Uncertainty	Units	Final N
Uncertainty of Width	μ_w	± 0.023	mm	10
Uncertainty of Thickness	μ_t	± 0.0015	mm	9

In (8)–(13), x_i is an individual measurement, $\bar{x}$ is the sample mean, S_x is the sample standard deviation, $S_{\bar{x}}$ is the standard error, $P_{\bar{x}}$ is the random uncertainty, t is the t value, and N is the number of retained measurements. Using this t was determined to be 2.262.

C. RSS Propagation of Area

The original cross-sectional area of the unknown specimen was calculated from the measured width and thickness. Since both quantities contained uncertainty, the uncertainty in area was determined by root-sum-square propagation (15). The area relation is equation 14.

$$A_o = wt \quad (14)$$

$$\mu_{A_o} = \left[\left(\mu_w \frac{\partial A_o}{\partial w}\right)^2 + \left(\mu_t \frac{\partial A_o}{\partial t}\right)^2\right]^{\frac{1}{2}} \quad (15)$$

D. RSS Propagation of Stress

The reported stress values for the unknown specimen were calculated by dividing the measured load by the original cross-sectional area. Since both load and area contained uncertainty, the uncertainty in stress was determined by RSS propagation using (1).

$$\mu_\sigma = \left[\left(\mu_P \frac{\partial \sigma}{\partial P}\right)^2 + \left(\mu_A \frac{\partial \sigma}{\partial A}\right)^2\right]^{\frac{1}{2}} \quad (16)$$

The propagated uncertainties for the reported stress-based properties of the unknown specimen are listed in Table IX.

TABLE IX
RSS PROPAGATION OF STRESS FOR UNKNOWN SPECIMEN

Name	Symbol	Value	Uncertainty	Units
.2% Offset Yield Strength	σ_y (.2%)	524.8	± 2.88	MPa
Visual Yield Strength	σ_y (Visual)	497.8	± 3.05	MPa
Ultimate Strength	σ_u	574.8	± 3.15	MPa
Breaking Strength	σ_b	322	± 1.77	MPa

E. *Monte Carlo Simulation for Uncertainty Analysis*

A Monte Carlo simulation was used to quantify the uncertainty in Young's modulus for the unknown specimen [5]. The chosen linear-elastic part of the specimen-1 data displayed in table X was used to run the simulation. Fig. 16 displays the resulting modulus value distribution.

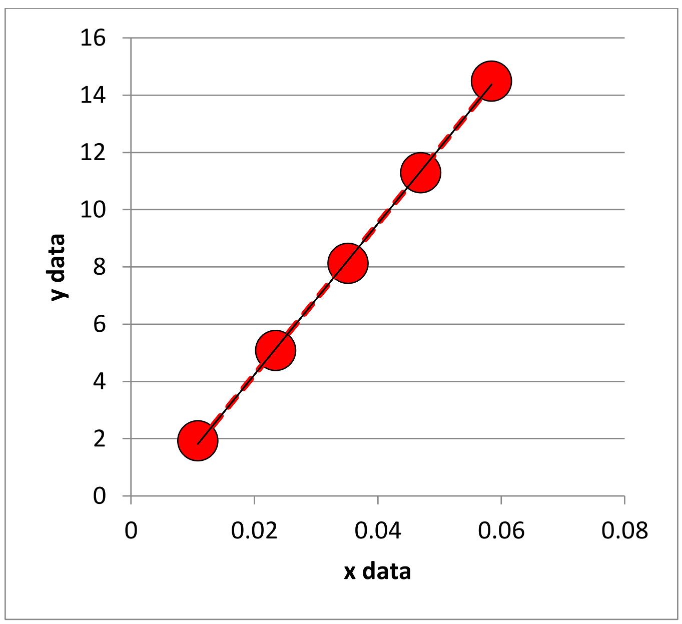

Fig 16. Monte Carlo histogram for Young's modulus of the unknown *specimen.*

TABLE X
MONTE CARLO MEASUREMENTS AND UNCERTAINTIES

Strain (mm/mm)	Stress (MPa)
$0.01085\ \pm 1\text{E}-7$	$1.9326\ \pm 0.0003$
$0.02345\ \pm 2.5\text{E}-7$	$5.07952\ \pm 0.00086$
$0.03520\ \pm 4.0\text{E}-7$	$8.1275\ \pm 0.0014$
$0.04696\ \pm 5.5\text{E}-7$	$11.293\ \pm 0.00191$
$0.05846\ \pm 7.5\text{E}-7$	$14.4905\ \pm 0.00245$

F. RSS Propagation of Density

The density of the unknown specimen was calculated from its measured mass and estimated volume using equation 17.

$$\rho = \frac{m}{V} \tag{17}$$

$$\mu_\rho = \left[\left(\mu_m \frac{\partial \rho}{\partial m}\right)^2 + \left(\mu_V \frac{\partial \rho}{\partial V}\right)^2\right]^{\frac{1}{2}} \tag{18}$$

G. RSS Propagation of Specific Strength

For the unknown metal, specific strength was based on the 0.2% offset yield strength because the specimen behaved as a ductile material using (6).

$$\mu_{STR} = \left[\left(\mu_\rho \frac{\partial STR}{\partial \rho}\right)^2 + \left(\mu_\sigma \frac{\partial STR}{\partial \sigma}\right)^2\right]^{\frac{1}{2}} \tag{19}$$

H. RSS Propagation of Specific Stiffness

RSS was applied to (5) to produce (20) to calculate the uncertainty in the specific stiffness.

$$\mu_{STI} = \left[\left(\mu_\rho \frac{\partial STI}{\partial \rho}\right)^2 + \left(\mu_E \frac{\partial STI}{\partial E}\right)^2\right]^{\frac{1}{2}} \tag{20}$$

I. RSS Propagation of Percent Elongation by Length

RSS was applied to (21) to get (22) in order to calculate the uncertainty in the percent elongation measured by length comparison.

$$\%EL_l = \frac{L_f - L_o}{L_o} \times 100 \tag{21}$$

$$\mu_{\%EL_l} = \left[\left(\mu_{L_f} \frac{\partial \%EL_l}{\partial L_f}\right)^2 + \left(\mu_{L_o} \frac{\partial \%EL_l}{\partial L_o}\right)^2\right]^{\frac{1}{2}} \tag{22}$$

J. RSS Propagation of Percent Elongation by Strain

RSS was applied to (23) to get (24) in order to calculate the uncertainty in the percent elongation as determined by strain comparison.

$$\rho = \frac{m}{V} \tag{23}$$

$$\mu_{\%EL_\epsilon} = \left[\left(\mu_{\epsilon_f} \frac{\partial \%EL_\epsilon}{\partial \epsilon_f}\right)^2 + \left(\mu_{\epsilon_y} \frac{\partial \%EL_\epsilon}{\partial \epsilon_y}\right)^2\right]^{\frac{1}{2}} \tag{24}$$

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